THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

Dec 21, 2024 2:30 p.m. – 5:00 p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL Questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (36 points) **Prove or disprove** each of the following statements:

   (a) For any sets A, B, C, $A \cup (B \triangle C) = (A \cup B) \triangle (A \cup C)$.

   (b) There exist a prime $p \geq 3$ such that $p^4 + 31$ is not a multiple of 16.

   (c) For any nonempty set A and for any partitions P_1 and P_2 of A, the set

   $$P = \{S \in \mathcal{P}(A) \setminus \{\emptyset\} : S = S_1 \cap S_2 \text{ for some } S_1 \in P_1 \text{ and } S_2 \in P_2\}.$$

   gives a partition of A.

   (d) There exist a finite nonempty set A and a bijective function $f : A \longrightarrow A$ satisfying $f \neq id_A$ and $f \circ f \neq id_A$ and $f \circ f \circ f = id_A$.

   (e) For any nonempty sets A, B with $A \cap B \neq \emptyset$, if $A \setminus B$ is numerically equivalent to $B \setminus A$, then A is numerically equivalent to B.

   (f) For any nonempty infinite sets A and B, if A is uncountable and there is a surjective function from A to B, then B is uncountable.

2. (14 points) In this question, you may use without proof the fact that

   $$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \quad \text{for any } n \in \mathbb{N}.$$

   (a) Prove by induction that

   $$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2 \quad \text{for any } n \in \mathbb{N}.$$

   (b) Use (a) to express 2025 as a sum of cubes.

   (c) Let $a_1, a_2, a_3, \ldots$ be a sequence of positive real numbers such that $a_1 = 1$ and

   $$a_1^3 + a_2^3 + \cdots + a_n^3 = (a_1 + a_2 + \cdots + a_n)^2 \quad \text{for any } n \in \mathbb{N}.$$

   Prove that the only sequence satisfying the above conditions is the sequence with $a_n = n$ for any $n \in \mathbb{N}$.

   (*Hint*: Let $P(n)$ be the statement $a_n = n$. Then prove by induction that $P(n)$ is true for any $n \in \mathbb{N}$.)

3. (14 points) In this question, you may use without proof the fact that the sum/product of two rational numbers is rational.

Denote by $\mathbb{I}$ the set of all irrational numbers. Let $f : \mathbb{I} \longrightarrow \mathbb{R}$ be the function defined by

$$f(x) = \frac{ax+b}{cx+d} \quad \text{for some } a, b, c, d \in \mathbb{Q} \text{ with } ad - bc \neq 0.$$

(a) Prove that $cx + d \neq 0$ for any $x \in \mathbb{I}$ (and hence $f(x)$ is well-defined for any $x \in \mathbb{I}$).

(b) Prove that f is injective.

(c) Find the range of f. Prove your answer.

4. (20 points) Let $f : \mathbb{N} \longrightarrow \mathbb{N}$ be a function satisfying

$$f(f(m)) = m + 2025 \quad \text{for any } m \in \mathbb{N}. \qquad (*)$$

(a) For any $m \in \mathbb{N}$, prove by induction that

$$f(m + 2025n) = f(m) + 2025n \quad \text{for any } n \in \mathbb{N}.$$

(b) Prove that for any $m, n \in \mathbb{N}$, if $m \equiv n \pmod{2025}$, then $f(m) \equiv f(n) \pmod{2025}$. (*Hint*: Use (a).)

(c) Let A be the set $\{0, 1, 2, \ldots, 2024\}$. Define a function $g : A \longrightarrow A$ such that for any $m \in A$, $g(m)$ is the remainder when $f(m)$ is divided by 2025.
Prove that $g(g(m)) = m$ for any $m \in A$.
(*Hint*: By the definition of g, we have $g(m) \equiv f(m) \pmod{2025}$ for any $m \in A$. Then use (b).)

(d) Prove that g has a fixed point, that is, $g(m_0) = m_0$ for some $m_0 \in A$.
(*Hint*: Consider the sets $\{0,\ g(0)\}, \ldots, \{2024,\ g(2024)\}$ and use (c).)

(e) Hence, or otherwise, prove that there does not exist a function $f : \mathbb{N} \longrightarrow \mathbb{N}$ satisfying $(*)$.

5. (16 points) Let A be a nonempty set and R be a relation on A. We say that R is *anti-symmetric* if for any $x, y \in A$, $(xRy$ and $yRx) \Longrightarrow x = y$.

We say that R is a *partial ordering* if it is reflexive, anti-symmetric and transitive.

(a) Let S be a relation on $\mathbb{R}^2$ defined by $(x_1, y_1)\, S\, (x_2, y_2)$ if $\sqrt{x_1^2 + y_1^2} \leq \sqrt{x_2^2 + y_2^2}$. **Prove or disprove** that S is a partial ordering.

(b) Let $B = \{(x, y) \in \mathbb{R}^2 : (x, y) = (0, 0) \text{ or } x > 0\}$. Let T be a relation on $\mathbb{R}^2$ defined by $(x_1, y_1)\, T\, (x_2, y_2)$ if $(x_1 - x_2, y_1 - y_2) \in B$. **Prove or disprove** that T is a partial ordering.

(c) **Prove or disprove** the statement: "*There exists a nonempty set A such that no relation on A can be both a partial ordering and an equivalence relation.*" The solution below is generated by an AI-based tool (with slight modification of wordings).

> "Proof": Consider a set A with at least two distinct elements. Assume to the contrary that there exists a relation R on A that is both a partial ordering and an equivalence relation.
>
> Since a partial ordering is anti-symmetric and an equivalence relation is symmetric, so R satisfies both anti-symmetric and symmetric properties. Now, suppose a, b are two distinct elements of A such that $(a, b) \in R$. Then $(b, a) \in R$ as R is symmetric. Since R is also anti-symmetric, this would imply that $a = b$ since both (a, b) and (b, a) are in R. However, this contradicts the assumption that a, b are distinct elements.
>
> Therefore, we have proved that there exists a nonempty set A such that no relation on A can be both a partial ordering and an equivalence relation.

If you think you should **prove** the given statement, comment and find out the mistake(s) (if any) in the AI-generated proof.

If you think you should **disprove** it, point out the mistake(s) in the "proof" and explain briefly what goes wrong. Then provide a correct disproof.

******* **END OF PAPER** *******